Immunization

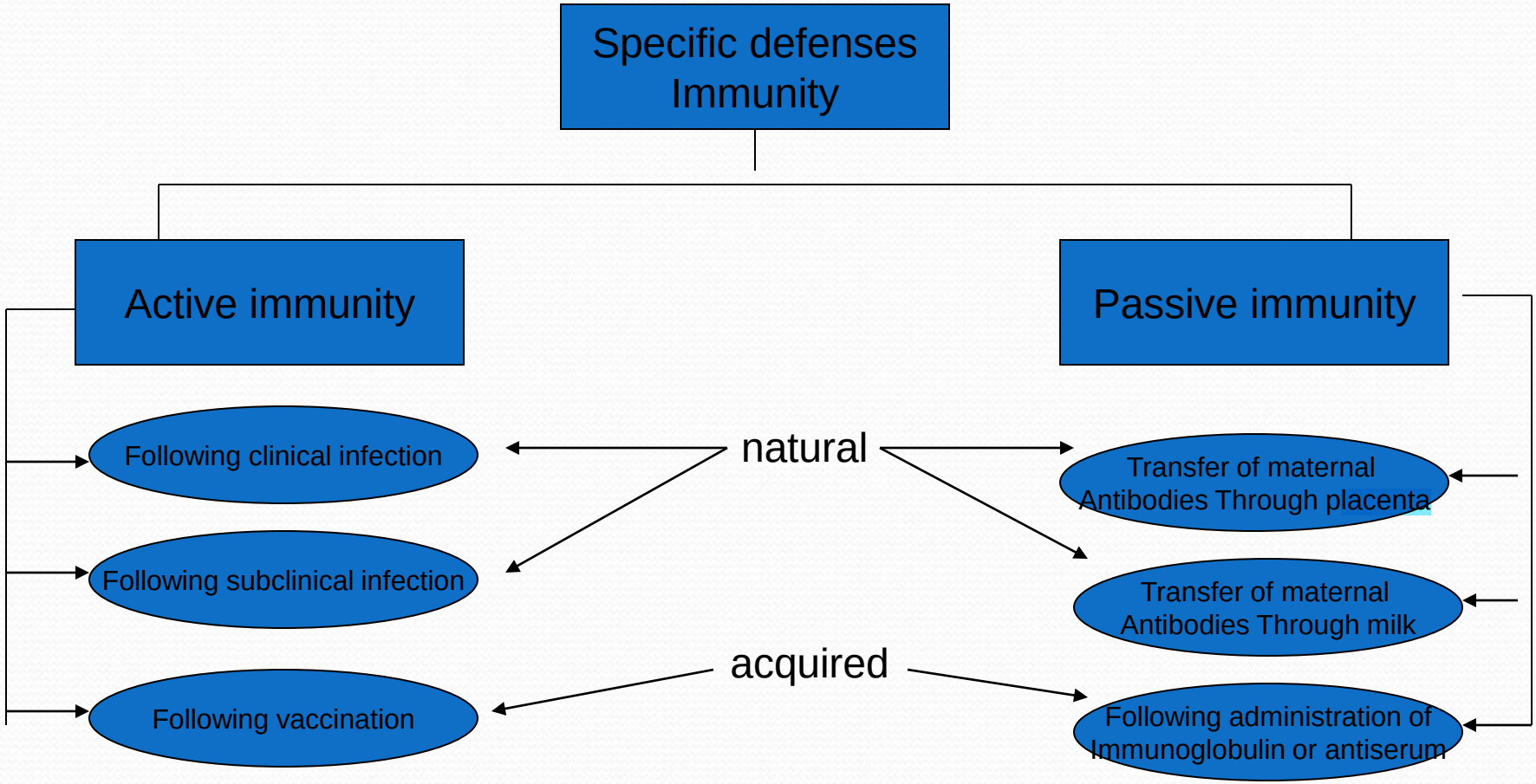
Active immunity

- Resistance developed in response to stimulus by an antigen (infecting agent or vaccine) and is characterized by the production of antibodies by the host.

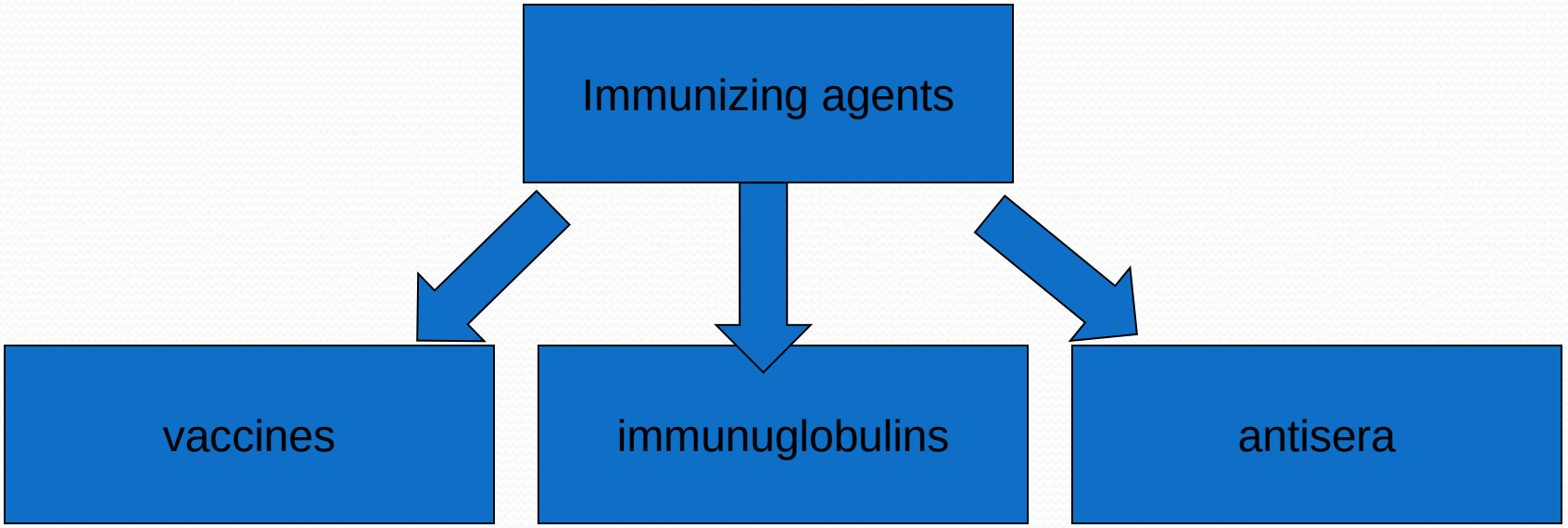
Passive immunity

- Immunity conferred by an antibody produced in another host. It may be acquired naturally or artificially (through an antibody-containing preparation).

Immunity


Specific defenses Immunity
Active immunity
Passive immunity
Following clinical infection
Following subclinical infection
Following vaccination
natural
acquired
Transfer of maternal Antibodies Through placenta
Transfer of maternal Antibodies Through milk
Following administration of Immunoglobulin or antiserum

Immunizing agents



Immunoglobulins

- There are 5 major classes: IgM, IgA, IgG, IgE, IgD.
- Two types of immunoglobulin preparations are available for passive immunization:
  - Normal human immunoglobulin
  - Specific (hyper-immune) human immunoglobulin

Antisera or antitoxins

- These are materials prepared in animals or non human sources such as horses.

Immunoglobulin and antiserum

Human normal immunoglobulin	Human specific immunoglobulin	Non human ig (antisera)
Hepatitis A Measles Rabies Tetanus Mumps	Hepatitis B Varicella Diphtheria	Diphtheria Tetanus Gas gangrene Botulism Rabies

Vaccination

- Vaccination is a method of giving antigen to stimulate the immune response through active immunization.

- A vaccine is an immuno-biological substance designed to produce specific protection against a given disease.
- A vaccine is “antigenic” but not “pathogenic”.

Types of vaccines

- Live vaccines
- Attenuated live vaccines
- Inactivated (killed vaccines)
- Toxoids
- Polysaccharide and polypeptide (cellular fraction) vaccines
- Surface antigen (recombinant) vaccines.

Live vaccines

- Live vaccines are made from live infectious agents without any modification.
- The only live vaccine is "Variola" small pox vaccine, made of live vaccinia cow-pox virus (not variola virus) which is not pathogenic but antigenic, giving cross immunity for variola.

Live attenuated (avirulent) vaccines

- Virulent pathogenic organisms are treated to become attenuated and avirulent but antigenic. They have lost their capacity to induce disease but retain their immunogenicity.
- Live attenuated vaccines should not be administered to persons with suppressed immune response as :
  - Leukemia and lymphoma
  - Other malignancies
  - Receiving corticosteroids and anti-metabolic agents
  - Radiation
  - pregnancy

Inactivated (killed) vaccines

- Organisms are killed or inactivated by heat or chemicals but remain antigenic. They are usually safe but less effective than live attenuated vaccines. The only absolute contraindication to their administration is a severe local or general reaction to a previous dose.

Toxoids

- They are prepared by detoxifying the exotoxins of some bacteria rendering them antigenic but not pathogenic. Adjuvant (e.g. alum precipitation) is used to increase the potency of vaccine.

Polysaccharide and polypeptide (cellular fraction) vaccines

- They are prepared from extracted cellular fractions e.g. meningococcal vaccine from the polysaccharide antigen of the cell wall, the pneumococcal vaccine from the polysaccharide contained in the capsule of the organism, and hepatitis B polypeptide vaccine.

- Their efficacy and safety appear to be high.

Surface antigen (recombinant) vaccines.

- It is prepared by cloning HBs Ag gene in yeast cells where it is expressed. HBs Ag produced is then used for vaccine preparations.

- Their efficacy and safety also appear to be high.

Types of vaccines

Live vaccines	Live Attenuated vaccines	Killed Inactivated vaccines	Toxoids	Cellular fraction vaccines	Recombinant vaccines
•Small pox variola vaccine	•BCG •Typhoid oral •Plague •Oral polio •Yellow fever •Measles •Mumps •Rubella •Intranasal Influenza •Typhus	•Typhoid •Cholera •Pertussis •Plague •Rabies •Salk polio •Intra-muscular influenza •Japanise encephalitis	•Diphtheria •Tetanus	•Meningococcal polysaccharide vaccine •Pneumococcal polysaccharide vaccine •Hepatitis B polypeptide vaccine	•Hepatitis B vaccine

Routes of administration

- subcutaneous
- intramuscular route (most vaccines)
- Oral route (sabin vaccine)
- Intra dermal route (BCG vaccine)
- Intranasal route (live attenuated influenza vaccine)

Scheme of immunization

- Primary vaccination
  - One dose vaccines (BCG, variola, measles, mumps, rubella, yellow fever)
  - Multiple dose vaccines (polio, DPT, hepatitis B)
- Booster vaccination

  To maintain immunity level as it declines after some time has elapsed (DT, MMR).

Periods of maintained immunity due to vaccines

- Short period (months): cholera vaccine
- Three to five years: DPT vaccine
- Five or more years: BCG vaccine
- Ten years: yellow fever vaccine
- Solid immunity: measles, mumps, and rubella vaccines.

Levels of effectiveness

- Absolutely protective(100%): yellow fever vaccine
- Almost absolutely protective (99%): Variola, measles, mumps, rubella vaccines, and diphtheria and tetanus toxoids.
- Highly protective (80-95%): polio, BCG, Hepatitis B, and pertussis vaccines.
- Moderately protective (40-60%) cholera vaccine, and influenza killed vaccine.

The Cold Chain

- The "cold chain" is a system of storage and transport of vaccines at low temperature from the manufacturer to the actual vaccination site.
- The cold chain system is necessary because vaccine failure may occur due to failure to store and transport under strict temperature controls.

The Cold Chain

- Among the vaccines, polio is the most sensitive to heat, requiring storage at minus 20 degree C.
- Vaccines which must be stored in the freezer compartment are : polio and measles.
- Vaccines which must be stored in the COLD PART but never allowed to freeze are : typhoid, DPT, tetanus toxoid, DT, BCG and diluents

HAZARDS OF IMMUNIZATION

- No immune response is entirely free from the risk of adverse reactions or remote squeal. The adverse reactions that may occur may be grouped under the following heads:

1. *Reactions inherent to inoculation*
2. *Reactions due to faulty techniques*
3. *Neurological involvement*
4. *Aggressive reactions*
5. Others

1. ***Reactions inherent to inoculation:***

These may be local or general reactions. The local reactions may be pain, swelling, redness, tenderness and development of a small nodule or sterile abscess at the site of injection.

- The general reactions may be fever, malaise, headache and other constitutional symptoms.
- Most killed bacterial vaccines (e.g., typhoid) cause some local and general reactions.
- Diphtheria and tetanus toxoids and live polio vaccine cause little reaction.

2. *Reactions due to faulty techniques:*

Faulty techniques may relate to

- faulty production of vaccine (e.g. inadequate inactivation of the microbe, inadequate detoxication),
- too much vaccine given in one dose,
- improper immunization site or route,
- vaccine reconstituted with incorrect diluents.
- vaccine or diluents contaminated,
- vaccine stored incorrectly,

- contraindications ignored (e.g. a child who experienced a severe reaction after a previous dose of DPT vaccine is immunized with the same vaccine),

- Use of improperly sterilized syringes and needles carry the hazard of hepatitis B virus, and staphylo - and streptococcal infection

- *3. Neurological involvement:*
- Neuritic manifestations may be seen after the administration of serum or vaccine. The well-known examples are the postvaccinial encephalitis and encephalopathy following administration of anti -rabies and smallpox vaccines.
- GuillainBarre syndrome in association with the swine influenza vaccine is another example.

4. Provocative reactions:

- Occasionally following immunization there may occur a disease totally unconnected with the immunizing agent (e.g., provocative polio after DPT or DT administration against diphtheria).
- The mechanism seems to be that the individual is harboring the infectious agent and the administration of the vaccine shortens the incubation period and produces the disease or what may have been otherwise only a latent infection is converted into a clinical attack.

5. *Others:*

- These may comprise damage to the fetus (e.g., with rubella vaccination);

The risk of adverse reactions can be reduced by

- proper sterilization of syringes and needles, by proper selection of the subject and the product, and if due care is exercised in carrying out the procedure. Measles and BCG vaccines should be reconstituted only with the diluent supplied by the manufacturer.
- Reconstituted vaccine should be discarded at the end of each immunization session and NEVER retained for use in subsequent sessions.

The risk of adverse reactions can be reduced by

- In the refrigerator of the immunization centre, no other drug and substances should be stored beside vaccines

- Training of immunization worker and their close supervision to ensure that proper procedures are being followed are essential to prevent complications and deaths following immunization.

Vaccination Coverage

- Vaccination coverage is the percent of at risk or susceptible individuals, or population who have been fully immunized against particular diseases by vaccines or toxoids. To be significantly effective in prevention of disease on mass or community level at least a satisfactory proportion (75% or more) of the at risk population must be immunized.

Ways of achieving satisfactory immunization coverage

- Efficient immunization service; urban and rural
- Health awareness and cooperation of the public
- Outreach programs in rural and nomad areas, and home visits

Application of active immunization

- Infants and children expanded immunization program (schedule)
- Active immunization for adult females
- Vaccination for special occupations
- Vaccination for special environmental situations

Obligatory Vaccines

Age	Vaccine	Route	Dose	Nature of vaccine
Up to 1 week	Polio	Oral	2-3 drops	Live attenuated (zero dose)
Up to 3 month	BCG	ID	0.1 ml	Live attenuated
2 Months	Polio DPT Hepatitis-B	Oral IM IM	2-3 drops 0.5 ml 0.5ml	First dose First dose First dose
4monthes	2nd dose for polio, DPT and Hepatitis B.			
6 months	3rd dose polio, DPT and HepatitisB.			
9months	Polio Vit. A 100,000 IU	oral oral	2-3 drops	4th dose.
12 months	Polio MMR	oral SC	2-3 drops 0.5 ml	5th dose Mealses,mumps,rubella.
18 months	Polio DPT MMR Vit. A 200,000 IU	Oral IM SC oral	2-3 drops 0. 5 ml 0.5ml	Booster dose. Booster dose 2nd dose .

جدول التطعيمات الأساسية

Basic vaccination schedule

VISIT	VACCINE	التطعيم	موعد الزيارة
AT Birh	B.C.G Hepatitis B	درن التهاب كبدي (ب)	عند الولادة
2Months	IPV DTP Hib Hepatitis B	شلل أطفال بالحقن ثلاثي بكتيري مستديمة نزلية التهاب كبدي (ب)	عمر شهرين
4Months	OPV DTP Hib Hepatitis B	شلل اطفال بالفم ثلاثي بكتيري مستديمة نزلية التهاب كبدي (ب)	عمري ٤ اشهر
6Months	OPV DTP Hib Hepatitis B	شلل اطفال بالفم ثلاثي بكتيري مستديمة نزلية التهاب كبدي (ب)	عمر ٦ اشهر
9Months	Measles	الحصبة	عمر ٩ اشهر
12Months	OPV MMR CHICKENPOX	شلل اطفال بالفم ثلاث فيروسي جدري مائي (عنقز)	عمر ١٢ شهر
18Months	OPV DTP Hib Hepatitis A	شلل اطفال بالفم ثلاثي بكتيري مستديمة نزلية التهاب كبدي (أ)	عمر ١٨ شهر
2 Years	Hepatitis A	التهاب كبدي (أ)	عمر ٢ سنة
4-6 Years	OPV DTP MMR	شلل اطفال بالفم ثلاثي بكتيري ثلاث فيروسي	من ٤ ـ ٦ سنوات

Basic Vaccination Schedule		جدول التطعيمات الأساسية	
Visit	Vaccine	اللقاح	الزيارة
At Birth	BCG	الدرن	الولادة
	HepB	الإلتهاب الكبدي (ب)	
months 2	IPV	شلل الأطفال المعطل	شهرين
	(DTP , HepB)	(الثلاثي البكتيري ؛ الإلتهاب الكبدي (ب) ؛ المستدمية النزلية)	
months 4	OPV	شلل الأطفال الفموي	4 شهور
	(DRP , HepB , Hib)	(الثلاثي البكتيري ؛ الإلتهاب الكبدي (ب) ؛ المستدمية النزلية)	
months 6	OPV	شلل الأطفال الفموي	6 شهور
	(DRP , HepB , Hib)	(الثلاثي البكتيري ؛ الإلتهاب الكبدي (ب) ؛ المستدمية النزلية)	
months 9	(Measles (mono	الحصبة المفرد	9 شهور
12 months	OPV	شلل الأطفال الفموي	12 شهر
	MMR	الثلاثي الفيروسي	
	Varicella	الجديري الماني	
18 months	OPV	شلل الأطفال الفموي	18 شهر
	(DTP m Hib)	(الثلاثي البكتيري ؛ المستدمية النزلية)	
	(Hepatitis (A	الإلتهاب الكبدي (أ)	
24 months	(Hepatitis (A	الإلتهاب الكبدي (أ)	24 شهر

Table 4-3Non obligatory vaccines in Egypt

Vaccine / Age	Route	2 months	4 months	6 months	12 months	15 months	18 months	19-23 months	2-3 Years	4-6 Years
Rotavirus	oral	Rota	Rota	Rota						
Haemophilus Influenzae type b	I.M	Hib	Hib	Hib	Hib					
pneumococcal	I.M	PCV	PCV	PCV	PCV					
Influenza	I.M				Influenza (Yearly)					
Varicella	S.C			varicella						Varicella
Hepatitis A	I.M				Hep A (2doses)					
Meningococcal	S.C								MP SV4	

Non obligatory vaccines in Egypt

Vaccine / Age	Route	2 months	4 months	6 months	12 months	15 months	18 months	19-23 months	2-3 Years	4-6 Years
Rotavirus (live att.)	oral	Rota	Rota	Rota						
Hemophilus Influenza type [B] (live att.)	I.M	Hib	Hib	Hib	Hib					
Pneumococcal (polysaccharid)	I.M	PCV	PCV	PCV	PCV					
Influenza (L.A)	I.M				Influenza (Yearly)					
Varicella (L.A)	S.C				Varicella					Varicella
Hepatitis A(LA)	I.M				Hep A (2doses)					
Meningococcal (polysacch.)	S.C								MP SV4	
HPV vaccine (inactivated)	IM	3 doses 0-2-6 from 11 to 18 years								

Active immunization for adult females

- MMR vaccine is given in adolescence before or after marriage, but not during pregnancy.

- Tetanus toxoid in pregnancy to prevent tetanus neonatorum in the newborn. In the first pregnancy on the third month and after 1 month. The third dose in the second pregnancy, and the fourth on the third pregnancy with a maximum of 5 doses. If 10 years elapse, and then pregnancy occurs, the doses are given from the start.

- Live attenuated vaccines should not be given during pregnancy.

Vaccination for special occupations

- Health care workers: hepatitis B, influenza, MMR, polio
- Public safety personnel (police, fire fighters) and staff of institutions for the developmentally disabled: hepatitis B, influenza
- Vets and animal handlers: rabies, plague and anthrax
- Sewage workers: DT, hepatitis A, polio,

- Military troops and camp dwellers: pneumococcal, meningococcal, influenza, BCG (for non reactors), tetanus

Vaccinations for special health status persons

- Immuno-compromised persons (Leukemia, lymphoma, HIV, malignancy…)
- Hemodialysis and transplantation
- Should receive the following vaccines according to their situation: HBV, Influenza, Pneuomococcal vaccines

Every vaccine you should comment on the following

- Type of vaccine
- Obligatory or not
- Schedules
- Route of administration
- Precaution and contraindication
- Side effects

BCG vaccine

- Live attenuated vaccine
- Given in the first 3 months
- Intradermal
- Side effects
  - Axillary lymphadenpathy
  - disseminated TB







Poliomyelities virus vaccine

- **Sabin vaccine (trivalent):**
  - **Given orally**
  - **Life long immunity**
  - **Herd immunity**
  - **Induce systemic and local intestinal immunity.**
  - **Fragile and need proper storage.**
- **Salk vaccine (monovalent):**
  - **Used in developed countries immunocomp.**
  - **Given I.M.**
  - **Expensive**
  - **Inactivated virus**
  - **Give only systemic immunity**





DPT Vaccine

- **Triple vaccine** **(Diphtheria toxoid – Tetanus toxoid – Killed acellular pertussis)**
- **Route :IM**
- **Side effects**
  - **Local reactions (generally erythema and induration with or without tenderness) are common**
  - **Mild systemic reactions such as fever, drowsiness, and anorexia occur frequently.**
  - **Neurological complications (convulsion – rarely encephalopathy)**

Contraindications for DPT vacc.

- Sever febrile illness
- During polio epidemics
- History of sever adverse reaction (convulsions – encephalopathy – anaphylaxis) and give only DT
- Children with progressive neurologic disease
- After 6 ys of age use only DT









MMR

- **Live attenuated (Measles – Mumps – Rubella)**
- **SC injection**
- **At 12 and 18 months.**
- **Booster dose 4-6 ys then 13 ys. & another premarital dose for females only**
- **Side effects :**
  - ✓ Fever, malaise, rash (10 days after vac.)
  - ✓ Activation of dormant T.B. focus due to transient inhibition of T-cell function so tuberculin test should be done for those with –ve BCG scar.
  - ✓ Encephalitis is very rare.

Measles



Mumps



Rubella



Hepatitis B vaccine

- **Recombinant DNA vaccine**
- **IM at 2-4-6 months and zero dose at birth**
- **Minimal side effects**

Rotavirus vaccine

- **Live attenuated**
- **Oral vaccine**



Anti-menningities vaccines

- **Haemphilus influenza type b**
  - **IM**
  - **3-4 doses**
- **Pneumococal vaccine**
  - **IM**
  - **3-4 doses**
- **Meningococcal polysascaride vaccive**
  - **SC**
  - **High risk group**
  - **After age of 2 years**

Varicella vaccine

- **Live attenuated**
- **Prevent chicken pox**
- **SC route**
- **2 doses**



- **Influenza virus vaccine**
  - Annually for children equal and more than 15 months.
  - For those with chronic diseases
  - IM route
  - Inactivated
- **Swine Flu (H1N1)vaccine**
  - Live attenated
  - IM route
- **Hepatitis A vaccine**
  - Inactivated vaccine
  - IM injections
  - Children 12-23 months
  - 2 doses with 6mo. apart



Human Papilloma Virus Vaccine

- **Recombinant DNA vaccine**
- **IM route**
- **3 doses at 0-2-6 intervals**
- **Given to females aged 11-12 years**
- **Minimal side effects**

THANK YOU